

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Tutorial 1: Solutions

Date: Aug 10, 2023

1. Evaluate $V_{AB} = V_A - V_B$ from the circuit shown in Figure 1.

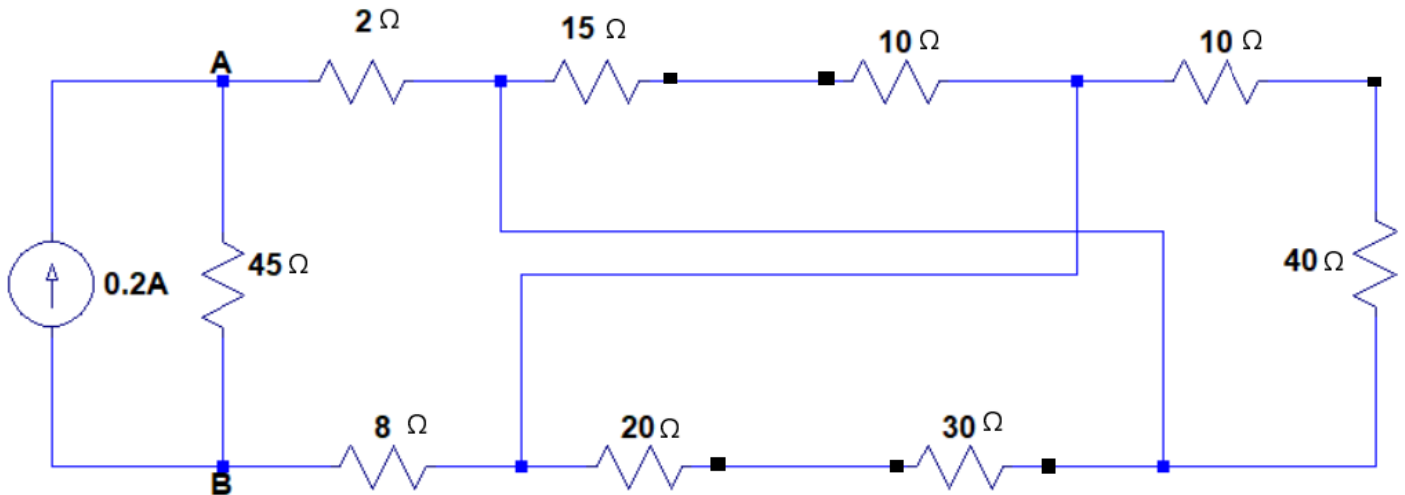


Figure 1: Evaluate V_{AB} from the above circuit.

Solution:

The given circuit nodes are labelled with $A, B, C \dots I, J, H$ and shown in Figure 2.

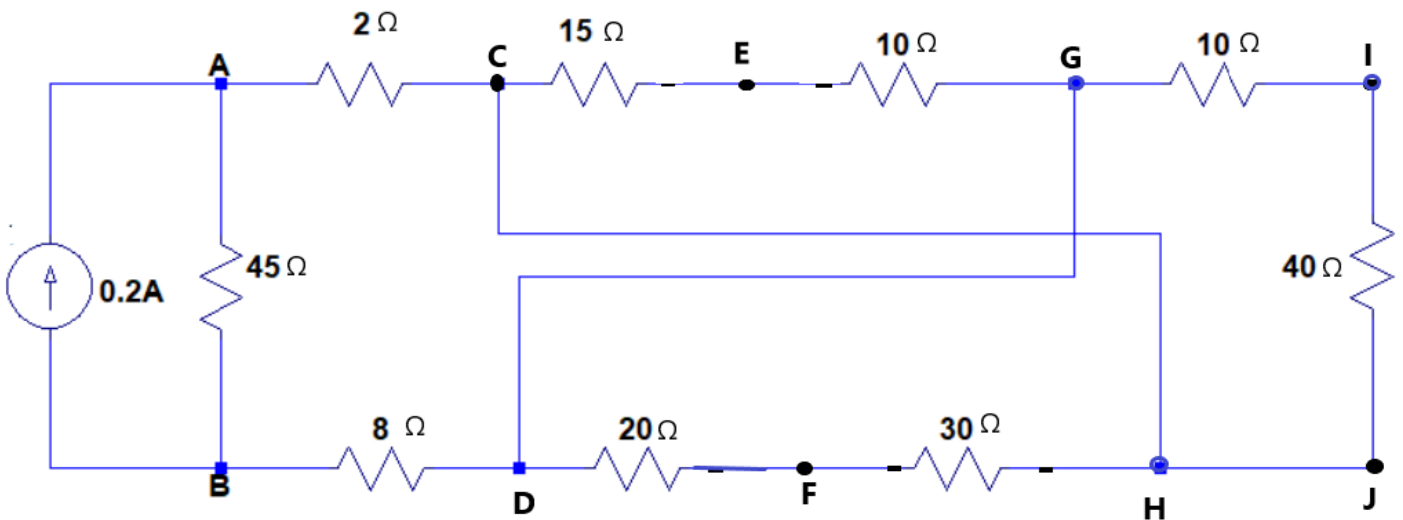


Figure 2:

The circuit shown in Figure 2 can be redrawn as shown in Figure 3.

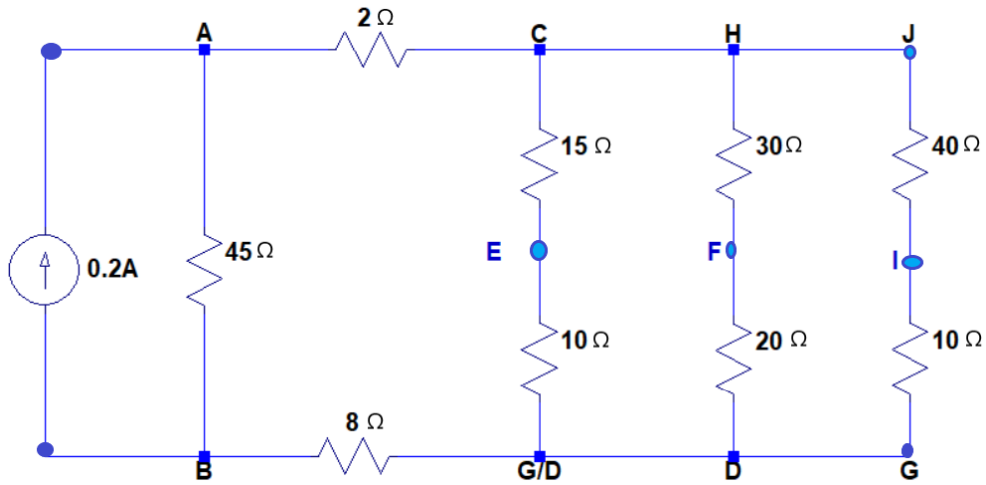


Figure 3:

The resistors in parallel in branches $C - G/D$ (25Ω), $H - D$ (50Ω) and $J - G$ (50Ω) yield an equivalent resistance of 12.5Ω (Figure 4).

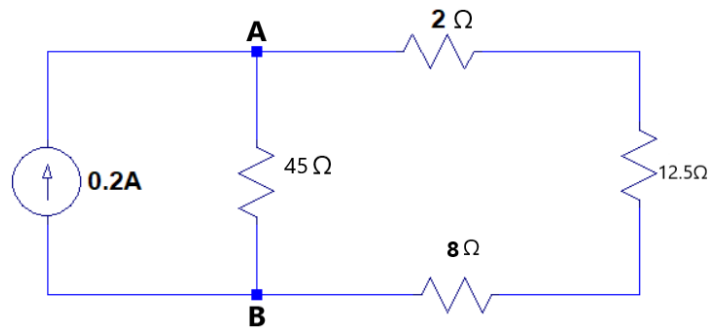


Figure 4:

Finally, the circuit is further simplified (Figure 5) to have an equivalent resistance of 15Ω ($45\Omega \parallel 22.5\Omega$).

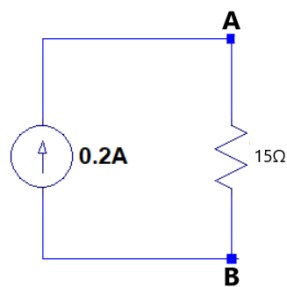


Figure 5:

Hence, $V_{AB} = 15\Omega \times 0.2A = 3V$.

2. Apply Ohm's Law, KCL and KVL to evaluate I_s , v_x and I_{in} from the circuit shown in Figure 6.

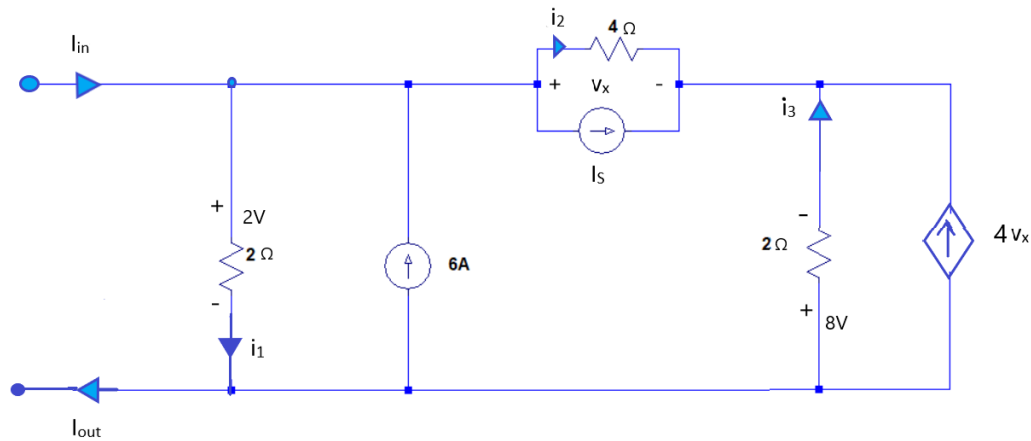


Figure 6: Evaluate I_s , v_x and I_{in} from the above circuit.

Solution:

The given circuit is redrawn in Figure 7.

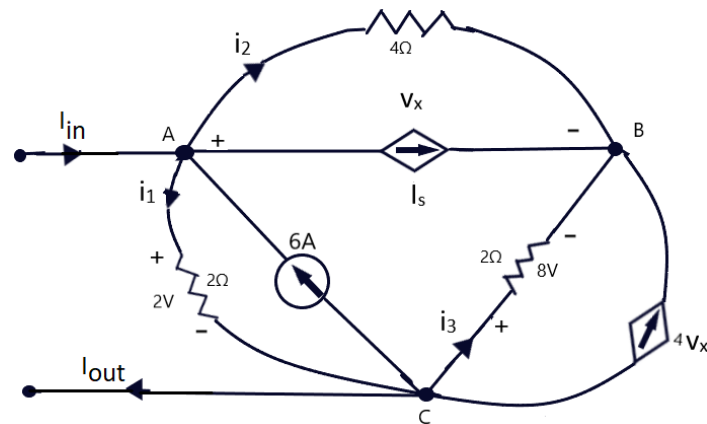


Figure 7:

$$i_1 = \frac{2V}{2\Omega} = 1A$$

$$i_3 = \frac{8V}{2\Omega} = 4A$$

KVL: A-B-C-A (voltage drop is taken positive)

$$v_x - 8 - 2 = 0 \Rightarrow v_x = 10V$$

Hence,

$$i_2 = \frac{10V}{4\Omega} = 2.5A$$

Apply KCL at Node B:

$$\begin{aligned} i_2 + I_s + i_3 + 4v_x &= 0 \\ \Rightarrow 2.5 + I_s + 4 + 40 &= 0 \end{aligned}$$

$$\Rightarrow I_s = -46.5A$$

Apply KCL at Node A:

$$I_{in} + 6 = i_1 + i_2 + I_s$$

$$\Rightarrow I_{in} + 6 = 1 + 2.5 - 46.5$$

$$\Rightarrow I_{in} = -49A$$

3. Obtain the truth table for the circuits shown below in the figure. Are they recognizable as any of the standard logic gates?

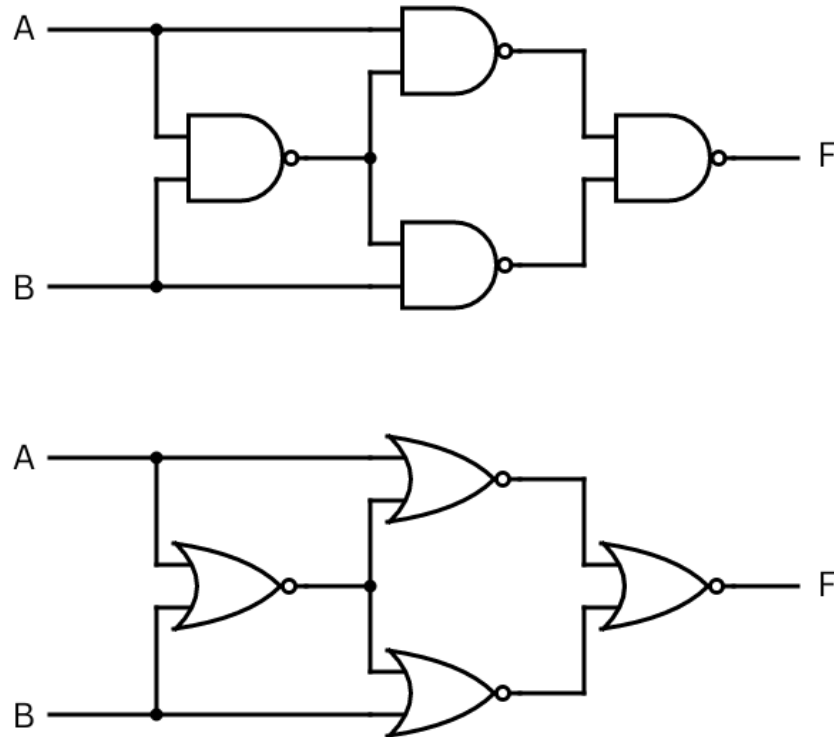
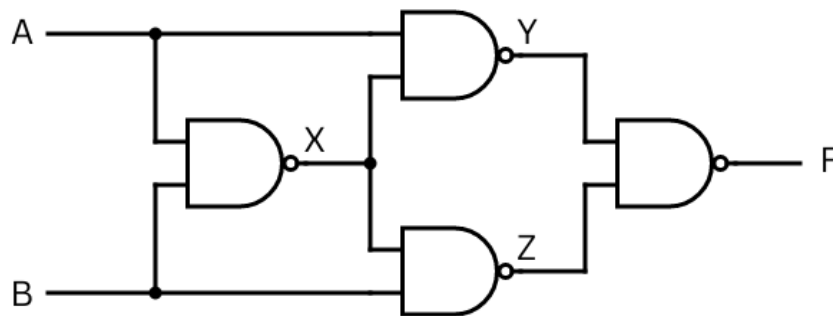


Figure 8: Examples of NAND logic and NOR logic

Solution: Consider the variables X , Y and Z shown below. Using the truth table of the 2-input NAND gate, we can compute what the values of X , Y and Z are as a function of the inputs A and B .

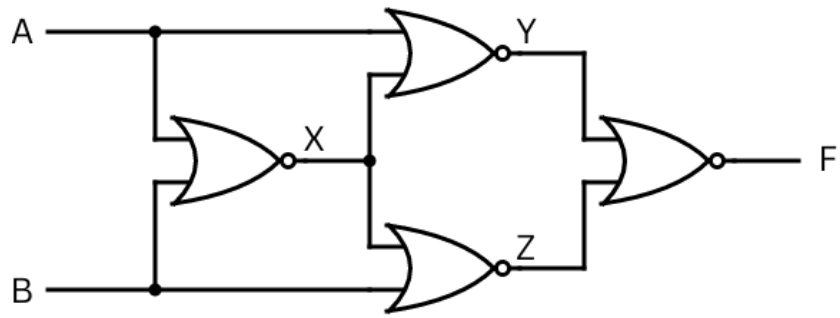


The truth table can be obtained as follows:

A	B	X	Y	Z	F
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	0

The circuit can be recognized as the XOR Gate from the truth table.

Similarly, consider the variables X , Y and Z shown below. Again, using the truth table of the 2-input NOR gate, we can compute what the values of X , Y and Z are as a function of the inputs A and B .

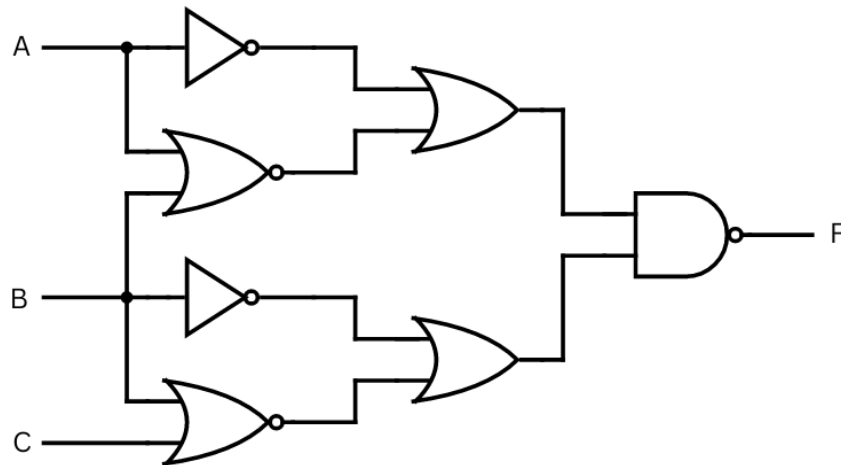


A	B	X	Y	Z	F
0	0	1	0	0	1
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

The circuit can be recognized as the XNOR Gate from the truth table.

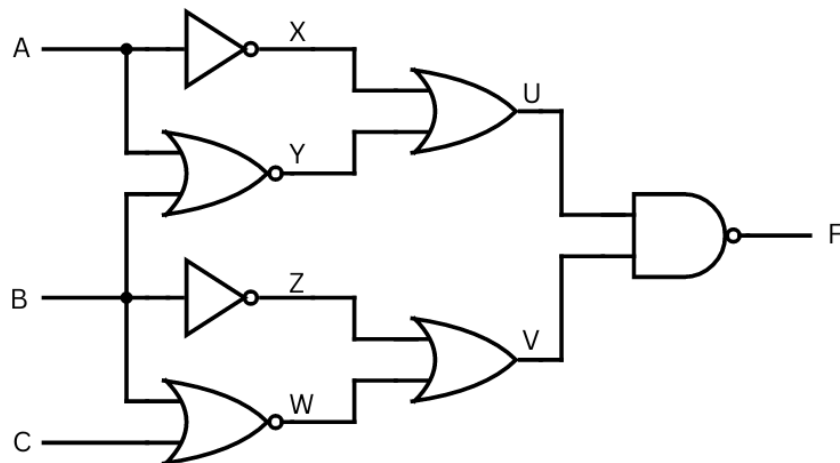
4. Consider the circuit shown below in the figure below.

- Obtain the truth table for the output F in terms of the inputs A , B and C .
- Obtain an algebraic expression for the output F in terms of the inputs A , B and C .



Solution: Consider the variables X , Y , Z , W , U and V shown below.

- Using the truth tables of the standard 2-input gates, we can compute what the values of X , Y , Z , W , U , V and F are as a function of the inputs A , B and C .



A	B	C	X	Y	Z	W	U	V	F
0	0	0	1	1	1	1	1	1	0
0	0	1	1	1	1	0	1	1	0
0	1	0	1	0	0	0	1	0	1
0	1	1	1	0	0	0	1	0	1
1	0	0	0	0	1	1	0	1	1
1	0	1	0	0	1	0	0	1	1
1	1	0	0	0	0	0	0	0	1
1	1	1	0	0	0	0	0	0	1

- Clearly, the following algebraic expressions hold.

$$\begin{aligned}
 X &= \overline{A} \\
 Y &= \overline{A + B} \\
 Z &= \overline{B} \\
 W &= \overline{B + C}
 \end{aligned}$$

$$\begin{aligned}
U &= X + Y = \overline{A} + \overline{A + B} \\
V &= Z + W = \overline{B} + \overline{B + C} \\
F &= \overline{U \cdot V} = \overline{(\overline{A} + \overline{A + B}) \cdot (\overline{B} + \overline{B + C})}
\end{aligned}$$

Alternative Solution

- When $A = 1$, we have $X = 0$ and $Y = 0$ irrespective of the values of B and C . Thus $U = 0$ and $F = 1$.
- Similarly, when $B = 1$, we have $Z = 0$ and $W = 0$ irrespective of the values of A and C . Thus $V = 0$ and $F = 1$.
- When A and B are both 0, we have $X = 1$ and $Z = 1$. This implies that $U = 1$ and $V = 1$ and thus $F = 0$ irrespective of the value of C .

The truth table is as below:

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

It may be noted that the circuit can be simplified as below:

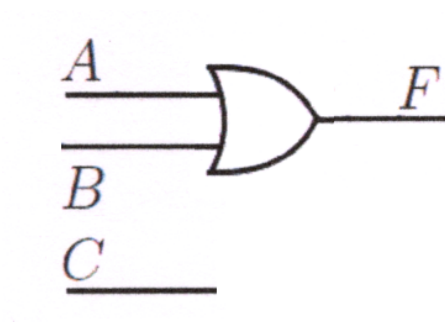


Figure 9: The simplified digital circuit